6.2 Properties of Transition Elements (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	6. Inorganic Chemistry (A Level Only)
Topic	6.2 Properties of Transition Elements (A Level Only)
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

This question is about transition metals.

Explain what is meant by a transition metal.

[1 mark]

Question 1b

i)

State the electronic configuration of Ti^{2+}.

[1]

ii)

Explain, using the electronic configuration, why zinc is found in the d-block of the periodic table but not classed as a transition metal.

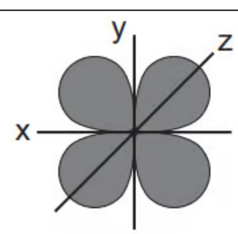
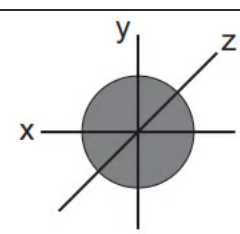
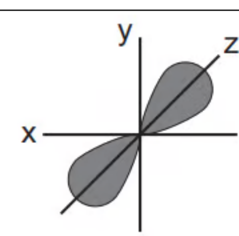
[1]

[2 marks]

Question 1c

Table 1.1 shows sketches of the shapes of some atomic orbitals.

Table 1.1

Shape of orbital			
Type of orbital			

Identify the type of orbital, s, p, or d.

[3 marks]

Question 1d

Transition metals have characteristic properties. One of these properties is that they form coloured ions.

State **two** other properties that transition metals or their ions exhibit.

[2 marks]

Question 2a

This question is about transition metal complexes.

Define the terms ligand and complex.

Ligand = ...

Complex = ...

[2 marks]

Question 2b

Transition metals can form complexes with different ligands.

Identify **one** species from the following list that does not act as a ligand. Explain your answer.

CO H_2O SCN^- H_2

[1 mark]

Question 2c

A complex contains one Co^{2+} ion, four ammonia molecules and two chloride ions.

i)
State the formula of this complex.

[1]

ii)
Name the shape of this complex.

[1]

[2 marks]

Question 2d

The H_2O ligands in $[Fe(H_2O)_6]^{3+}$ can be exchanged for other ligands.

Predict the shape of the complex ions formed after the following substitutions.

i)
All the H_2O ligands are exchanged for OH^- ligands.

[1]

ii)
The six H_2O ligands are exchanged for four Cl^- ligands.

[1]

[2 marks]

Question 2e

Cisplatin can be used to treat some types of cancer.

It is a square planar transition metal complex with a central platinum(II) ion, two chloride ligands and two ammonia ligands.

Complete the three-dimensional diagram in Fig. 2.1 to show the shape of cisplatin. Label and state the value of one bond angle.

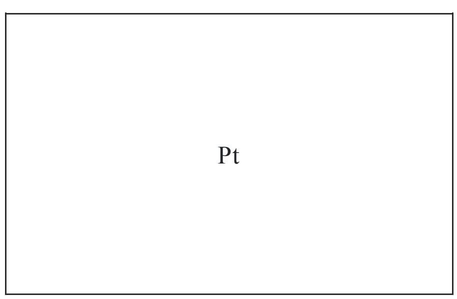


Fig. 2.1

[2 marks]

Question 3a

The d-orbitals of a transition metal are involved in the formation of complexes.

In terms of energy, explain what is meant by degenerate orbitals.

[1 mark]

Question 3b

Fig. 3.1 shows the shape of the d_{xy} orbital.

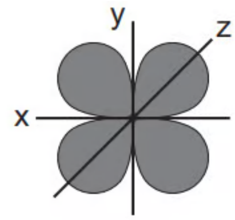


Fig. 3.1

The d_{xz} orbital has a similar shape but the lobes are between the x-axis and z-axis and the d_{yz} orbital has a similar shape but the lobes are between the y-axis and z-axis.

i)

State the difference between the shape of a d_{xy} orbital and the shape of a $d_{x^2-y^2}$ orbital.

[1]

ii)

Complete the three-dimensional diagram in Fig. 3.2 to show the shape of a d_{z^2} orbital.

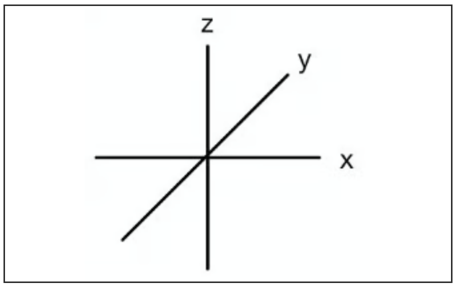


Fig. 3.2

[2]

[3 marks]

Question 3c

Suggest what causes d-orbitals in a transition metal ion to change from degenerate to non-degenerate.

[1 mark]

Question 3d

The splitting of degenerate d-orbitals gives two sets of non-degenerate d-orbitals at different energy levels.

Complete Table 3.1 to show the different energy levels of the non-degenerate d-orbitals in octahedral and tetrahedral complexes.

Table 3.1

	Higher energy	Lower energy
Octahedral		
Tetrahedral		

[2 marks]

Model Answers: Easy

1a

a) A transition metal is:

- A metal / element which forms one or more stable ions with an incomplete / partially filled d-orbital / d-subshell; [1 mark]

[Total: 1 mark]

- Transition metal chemistry has several key terms that you need to be able to define, including transition metal / element

1b

b)

i) The electronic configuration of Ti^{2+} is:

- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$
  OR
  $[Ar]3d^2$; [1 mark]

ii) Zinc is not classed as a transition metal because:

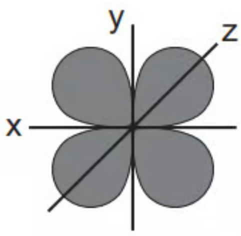
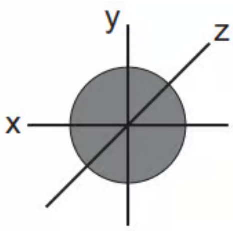
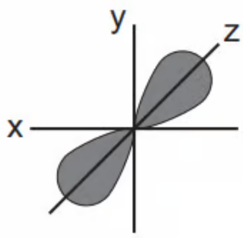
- Zinc(II) ion has a complete 3d subshell
  AND
  The electronic configuration of the zinc(II) ion is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$; [1 mark]

[Total: 2 marks]

- A transition metal is defined as an element that has an incomplete d subshell either in the element or in one of its ions, therefore:
  - Scandium(III) ion has no electrons in its d subshell
  - Zinc(II) ion has the electronic configuration of $[Ar]3d^{10}$, meaning the d subshell is complete and is also not classified as a transition metal

1c

c) The completed table is:

Shape of orbital			
Type of orbital	d; [1 mark]	s; [1 mark]	p; [1 mark]

[Total: 3 marks]

- You need to be able to identify and draw atomic orbitals
- You should be able to further identify:
  - The d orbital as d_{xy} as the lobes are in the xy-plane
  - The p orbital as p_z as the lobes are on the z-axis

1d

d) Other properties of transition metals and their complexes are:

Any **two** of the following:

- Form complexes / complex ions; [1 mark]
- Variable oxidation states; [1 mark]
- Behave as catalysts; [1 mark]

[Total: 2 marks]

- You need to know the characteristic properties of transition metals
- Exam questions will often go into further detail about these characteristic properties

2a

a) The definitions of the terms are:

- Ligand = a molecule or ion that has one or more lone pairs of electrons; [1 mark]

- Complex = a molecule or ion formed by a central metal atom or ion surrounded by one or more ligands; [1 mark]

[Total: 2 marks]

- You should know the definitions of these terms

2b

b) One species which is not a ligand is:

- H_2 / Hydrogen; [1 mark]
- No lone pair of electrons; [1 mark]

[Total: 2 marks]

- Ligands are ions or molecules which donate a lone pair of electrons to the central metal ion and form a coordinate bond
- Water (H_2O), thiocyanate (SCN^-) and carbon monoxide (CO) all have lone pairs of electrons which means that they can act as ligands in a complex

2c

c)

i) The formula of this complex is:

- $Co(NH_3)_4Cl_2$; [1 mark]

ii) The shape of this complex is:

- Octahedral; [1 mark]

[Total: 2 marks]

- **Remember:** Each chloride / Cl^- ligand has a single negative charge and each ammonia / NH_3 ligand has no charge
- The oxidation state of the cobalt ion is given as Co^{2+}
- Therefore, the overall charge on the complex must be 0
  - $+2 - 2 = 0$
- The central metal has 6 ligands surrounding it, which means that the shape must

be octahedral

2d

d) The shape of the complex ions formed after the following substitutions are:

i) All the H_2O ligands being exchanged for OH^- ligands results in:

- (An) octahedral (shape); [1 mark]

ii) The six H_2O ligands being exchanged for four Cl^- ligands results in:

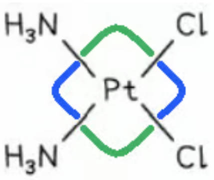
- (A) tetrahedral (shape); [1 mark]

[Total: 2 marks]

- If all the H_2O ligands are replaced with OH^- ligands this means there are still 6 coordinate bonds to the central metal ion
  - This results in an octahedral complex
- If six H_2O ligands are replaced with four Cl^- ligands this means the coordination number changes from 6 to 4
  - This results in a tetrahedral complex

2e

e) The completed three-dimensional structure and bond angle of cisplatin are:



- Correct structure
  AND
  No charge; [1 mark]
- Any bond angle identified
  AND
  Labelled as 90 ^{o}; [1 mark]

[Total: 2 marks]

- The question tells you that the complex is a cis-isomer
  - This means that the ammonia ligands are on the same side / next to each other and the chloride ligands are on the same side / next to each other
- The central platinum ion has a 2+ charge
- There are two chloride ligands, each with a 1- charge
- There are two ammonia ligands with no charge
  - This means that the overall charge is 0
- Since the complex has a square planar shape, there is no need for dot or wedge bonds because all of the bonds are in the same plane
- The square planar shape should automatically tell you that the bond angle is 90 º

3a

a) In terms of energy, degenerate orbitals are:

- Orbitals that are all at the same energy level
  OR
  Orbitals that are equal in energy; [1 mark]

[Total: 1 mark]

- Although this question is unlikely in an actual exam, you do need to not about degenerate and non-degenerate orbitals

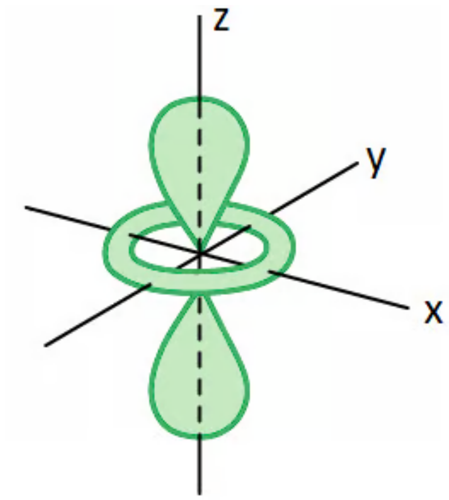
3b

b)

i) The difference between the shape of a d_{xy} orbital and the shape of a $d_{x^2-y^2}$ orbital is:

- The lobes of a d_{xy} orbital are <u>between</u> the x and y axes
  AND
  The lobes of a $d_{x^2-y^2}$ orbital are <u>on</u> the x and y axes; [1 mark]

ii) The three-dimensional diagram to show the shape of a d_{z^2} orbital is:



- Two lobes on the z-axis; [1 mark]
- Hoop / doughnut shape in the xy-plane; [1 mark]

[Total: 3 marks]

- It is more common to be asked to draw the shapes of different orbitals or identify the orbital from a diagram
  - But, if you are able to describe the difference (as seen in part (i)), then you have a clear understanding of orbital shapes

3c

c) The change of d-orbitals in a transition metal ion from degenerate orbitals to non-degenerate orbitals is caused by:

- Dative (covalent) / coordinate bonding to ligands
  OR
  Ligands causing the transition metal ion to not be isolated; [1 mark]

[Total: 1 mark]

- The splitting of degenerate d-orbitals into non-degenerate d-orbitals is caused by the dative covalent / coordinate bonding of a ligand
- The number / type / size of the ligand also affects how the degenerate d-orbitals split

3d

d) The completed table is:

	Higher energy	**Lower energy**
Octahedral	$d_{x^2-y^2}$ **AND** d_{z^2}	d_{xy} **AND** d_{xz} **AND** d_{yz}
	d	

[Total: 3 marks]

- It is more common to be asked to draw the shapes of different orbitals or identify the orbital from a diagram
    - But, if you are able to describe the difference (as seen in part (i)), then you have a clear understanding of orbital shapes

3c

c) The change of d-orbitals in a transition metal ion from degenerate orbitals to non-degenerate orbitals is caused by:

- Dative (covalent) / coordinate bonding to ligands
  OR
  Ligands causing the transition metal ion to not be isolated; [1 mark]

[Total: 1 mark]

- The splitting of degenerate d-orbitals into non-degenerate d-orbitals is caused by the dative covalent / coordinate bonding of a ligand
- The number / type / size of the ligand also affects how the degenerate d-orbitals split

3d

d) The completed table is:

	Higher energy	**Lower energy**
Octahedral	$d_{x^2-y^2}$ **AND** d_{z^2}	d_{xy} **AND** d_{xz} **AND** d_{yz}
Tetrahedral	d_{xy} **AND** d_{xz} **AND** d_{yz}	$d_{x^2-y^2}$ **AND** d_{z^2}

- All d-orbitals correctly assigned for octahedral; [1 mark]
- All d-orbitals correctly assigned for tetrahedral;

[Total: 2 marks]

- You only need to remember the energy levels of either the octahedral non-degenerate d-orbitals or the tetrahedral non-degenerate d-orbitals because they are the opposite of each other